

Ghassen Bargougui

Application Engineer | Java Applications, Cloud & Platform Engineering

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Profile

Application Engineer with seven years of Java and full-stack experience in banking, insurance, and card payments across Tunisia, Luxembourg, and Japan. Build and modernize services, design APIs, and deliver cloud deployments. Combine hands-on development with performance analysis, delivery automation, and coordination across engineering teams.

Professional experience

Application Engineer - Rakuten · Fukuoka, Japan | Dec 2025 - Present

Develop financial applications and cloud infrastructure. Help lead cloud migration alongside my manager and project manager, coordinating across engineering teams.

- Modernized Java services, resolved dependency and compatibility issues, updated tests, and documented the handover.
- Design and validate cloud deployments, including networking, resource configuration, and fallback plans, with infrastructure and security teams.
- Run performance tests with k6 and JMeter, investigate bottlenecks, and improve application design or resource allocation based on the results.
- Improve structured application logging and trace handling to support monitoring and troubleshooting.
- Develop Java APIs and service integrations, including session management and communication between application components.
- Automated local builds and deployments to make development environment setup repeatable and easier for teammates.
- Build CI/CD pipelines and work with partner teams to resolve deployment and connectivity issues, review designs, and support reliable delivery.

Software Engineer - Sogeti · Client assignment at BGL BNP Paribas · Luxembourg | Feb 2025 - Nov 2025

Developed and maintained retail-banking applications on assignment at BGL BNP Paribas.

- Implemented customer mandate workflows, from Spring Batch processing to Angular Material screens, with Spring Boot services.
- Developed REST APIs and persistence layers with Hibernate, PostgreSQL, and Oracle.
- Automated builds and deployments with Maven and Jenkins; maintained unit tests, SonarQube checks, and peer code reviews.

Java Full-Stack Engineer - CBTW · Luxembourg | Sep 2024 - Jan 2025

Developed full-stack web applications and modernized Java services using Spring Boot, Angular, and PostgreSQL.

- Refactored monolithic application components into microservices.
- Built and documented REST APIs for service integration, including API versioning and security requirements.
- Automated deployments with Jenkins and Docker, reducing manual release steps.

Analyst Developer - VERMEG for Banking and Insurance · Tunis, Tunisia | Jan 2023 - Aug 2024

Software Developer - VERMEG for Banking and Insurance · Tunis, Tunisia | Oct 2019 - Dec 2022

Progressed from Software Developer to Analyst Developer, combining backend development for insurance and financial applications with requirements analysis and solution design.

- Translated business requirements into backend designs and API contracts with business, architecture, and framework teams.
- Improved API response times through query optimization and caching.
- Documented solutions with UML and sequence diagrams and coordinated design decisions across teams.
- Implemented authentication and role-based access control with Keycloak; maintained CI/CD pipelines and rollback procedures.

Selected company projects

- **Insurance Workflow Modernization** (2023 - 2024): Led the migration of insurance workflows to a Camunda BPM-based architecture to reduce manual processing. Integrated ActiveMQ to automate bulk-processing workflows.
- **Insurance Product Configuration** (2022 - 2023): Profiled Java execution and memory usage with JProfiler and optimized the identified bottlenecks.

Technical skills

Java Application Engineering: Java, Spring Boot, Helidon, MicroProfile, Hibernate, RESTful API Design, OpenAPI, Maven

Cloud Delivery & CI/CD: Kubernetes, Google Cloud Platform (GCP), Google Kubernetes Engine (GKE), Docker, GitHub Actions, Jenkins

Performance & Observability: k6, JMeter, JProfiler, Cloud Logging, Log4j2

Data & Messaging: PostgreSQL, Oracle (PL/SQL), Redis, Kafka / Confluent, ActiveMQ

Frontend, Mobile & Browser: Angular, TypeScript, Vue.js, Nuxt, Node.js, Flutter, Dart, Chrome Extensions (MV3)

AI & Agent Tooling: Model Context Protocol (MCP), LLM API Integration

Quality & Security: JUnit, Mockito, SonarQube, Keycloak (OAuth2, OIDC, JWT), Git

Personal projects

Slide Agent · Creator & Maintainer · Open source · npm package | 2026

Open-source toolkit for generating editable PowerPoint presentations from code and AI tools.

- Built a TypeScript engine that generates native PowerPoint charts, tables, themes, and embedded fonts from structured layouts.
- Published @slide-agent/core on npm with a CLI, library API, and MCP server; added CSV-based template filling.

Technologies: TypeScript, Node.js, MCP, OOXML, LLM APIs, Vitest, GitHub Actions · slide-agent.ghassen.io · github.com/ghassenbrg/slide-agent

Pockito · Creator & Full-Stack Engineer · Personal project | 2025 - 2026

Personal and shared finance app with mobile and web clients.

- Built Flutter and Nuxt clients backed by Spring Boot, sharing business logic across REST and MCP interfaces.
- Built Kubernetes deployment and rollback automation, with automated tests and end-to-end release checks.

Technologies: Java, Spring Boot, Flutter, Nuxt, PostgreSQL, Redis, Keycloak, Kubernetes, MCP · pockito.ghassen.io

SubMate · Creator & Maintainer · Open source · Chrome extension | 2026

Chrome extension for subtitle translation on Netflix, TVer, and Prime Video, with on-device and optional cloud translation.

- Implemented on-device translation and local caching with Chrome's Translator API, without a dedicated backend or user accounts.
- Added optional cloud providers using the user's API key and localized the interface in English, Japanese, French, and Arabic, with right-to-left support.

Technologies: TypeScript, Chrome Extensions (MV3), Translator API, LLM APIs, IndexedDB, Vitest · submate.ghassen.io · github.com/ghassenbrg/SubMate

Other personal projects

- **Orbit Ways:** Browser-based multiplayer strategy board game with real-time game state over WebSockets. ghassen.io/orbitways
- **Traffic Forward** (2026): Spring Boot HTTP forwarding tool with configurable routing and request/response diagnostics. github.com/ghassenbrg/traffic-forward

Education, certification & languages

Engineering Degree (Diplôme d'ingénieur) in Software Architecture · ESPRIT (Private Higher School of Engineering and Technology), Tunis, Tunisia · Sep 2019 - Jul 2023

Bachelor's Degree in Computer Science · ISIMM (Higher Institute of Computer Science and Mathematics), Monastir, Tunisia · Sep 2015 - Jun 2019

Oracle Certified Professional, Java SE 11 Developer · Oracle · Sep 2022 · Oracle CertView: bit.ly/ocp11gb

Languages: Arabic (Native), English (Fluent), French (Fluent), Japanese (Basic)